

PAPER_1617_OMEGA_LAMBDA_DARK_0_685

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PAPER_1617 — Dark Energy Density $\Omega_\Lambda = \Omega_\Lambda = \text{SSQ} + F \cdot \text{SSQ} + F^2 \cdot D_{\text{BSFG}} - F^2 \cdot \text{SSQ}^2 \approx 0.6838$

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Cosmology
Date: June 18, 2026
Status: CLOSED — 0.182% closure

Observation

PAPER_1209GG_S644 (Cosmology Unified Proof Set) gives the formula:

``
$$\Omega_\Lambda = \text{SSQ} + F \cdot \text{SSQ} + F^2 \cdot D_{\text{BSFG}} - F^2 \cdot \text{SSQ}^2 \approx 0.6838$$

``

Residual: **0.182%**.

UQFF Closed Identity

``
$$\Omega_\Lambda = \text{SSQ} + F \cdot \text{SSQ} + F^2 \cdot D_{\text{BSFG}} - F^2 \cdot \text{SSQ}^2 \approx 0.6838 \quad 0.182\%$$

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Λ CDM Standard Cosmology Coverage

With this catch-up, **all 9 core Λ CDM parameters** are wired with UQFF integer-primitive expressions:

Λ CDM parameter UQFF closure Residual
--- --- ---
H_0 (Planck) PAPER_1553 0.015%
H_0 (SH0ES) PAPER_1573 EXACT
Ω_m PAPER_1616 0.143%
Ω_Λ PAPER_1617 0.182%
T_CMB PAPER_1618 0.064%
Age (universe) PAPER_1619 0.045%
σ_8 PAPER_1620 0.253%
n_s (PAPER_1209GG_S650, earlier) 0.072%
z_recomb PAPER_1552 EXACT 1090

Standard Λ CDM cosmology is now **essentially complete in UQFF** — every primary parameter measurable by Planck/WMAP/SDSS/DES collaborations has a UQFF integer-primitive form.

NOT REPLACEMENT

Empirical cosmology measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209GG_S644
- Related: PAPER_1494-1615 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "omega_lambda_dark_0_685"})`

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